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Unit 3 : Session 1 : SLO 1 — State and Prove De Morgan's Theorems

■ 1. State De Morgan's Theorems (Remember)

- The complement of the sum of two variables is equal to the PRODUCT of the COMPLEMENTS of the variables. $(A + B)' = A' \cdot B'$
- The complement of the product of two variables is equal to the SUM of the COMPLEMENTS of the variables. $(A \cdot B)' = A' + B'$

■ 2. Match the Expressions (Understand)

1. $(X + Y)'$ → d. The complement of a sum
2. $X' \cdot Y'$ → c. The product of the complements
3. $(X \cdot Y)'$ → b. The complement of a product
4. $X' + Y'$ → a. The sum of the complements

■ 3. Complete the Proof (Apply) — Truth table for $(A + B)' = A' \cdot B'$

- A=0, B=0: $A+B=0$, $(A+B)'=1$, $A'=1$, $B'=1$, $A' \cdot B'=1 \rightarrow \text{LHS} = \text{RHS}$
- A=0, B=1: $A+B=1$, $(A+B)'=0$, $A'=1$, $B'=0$, $A' \cdot B'=0 \rightarrow \text{LHS} = \text{RHS}$
- A=1, B=0: $A+B=1$, $(A+B)'=0$, $A'=0$, $B'=1$, $A' \cdot B'=0 \rightarrow \text{LHS} = \text{RHS}$
- A=1, B=1: $A+B=1$, $(A+B)'=0$, $A'=0$, $B'=0$, $A' \cdot B'=0 \rightarrow \text{LHS} = \text{RHS}$
- Since the columns for $(A + B)'$ and $A' \cdot B'$ are identical in every row, the theorem is proved.

■ 4. Simplify the Expression (Analyze)

- $F = ((A \cdot B)' + C)'$
- Step 1: $F = ((A \cdot B)')' \cdot C'$ (De Morgan's theorem applied to the outer complement)
- Step 2: $F = (A \cdot B) \cdot C'$ (law of double complementation)
- Final simplified expression: $F = A \cdot B \cdot C'$

■ 5. Construct a Proof (Create) — Truth table for $(A \cdot B)' = A' + B'$

- A=0, B=0: $A \cdot B=0$, $(A \cdot B)'=1$, $A'=1$, $B'=1$, $A'+B'=1 \rightarrow \text{LHS} = \text{RHS}$
- A=0, B=1: $A \cdot B=0$, $(A \cdot B)'=1$, $A'=1$, $B'=0$, $A'+B'=1 \rightarrow \text{LHS} = \text{RHS}$
- A=1, B=0: $A \cdot B=0$, $(A \cdot B)'=1$, $A'=0$, $B'=1$, $A'+B'=1 \rightarrow \text{LHS} = \text{RHS}$
- A=1, B=1: $A \cdot B=1$, $(A \cdot B)'=0$, $A'=0$, $B'=0$, $A'+B'=0 \rightarrow \text{LHS} = \text{RHS}$
- The columns for $(A \cdot B)'$ and $A' + B'$ are identical, so the second theorem is proved.